18 TECHNOLOGIES WE NEED RIGHT NOW. PLEASE.

Few things are more frustrating than bleeding edge technologies demonstrated in labs that have not found their way to the market yet.

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01 ARTIFICIAL PHOTOSYNTHESIS

Plants convert the carbon dioxide into glucose using water and sunlight. The glucose acts as their source of energy, and as a result of the reaction, oxygen is released into the atmosphere. The actions of millions of photosynthetic organisms pumped oxygen into the atmosphere about 4.5 billion years ago, during what is known as the great oxygenation event. Over the course of the industrial revolution, humans too have pumped the atmosphere with greenhouse gases, including carbon dioxide. The negative effects of this indiscriminate dumping is being seen all over the planet, and is projected to accelerate in the future. Labs around the world are trying to come up with catalysts that can mimic the action of photosynthesis in plants, to produce carbohydrates that can be a source of environmentally friendly fuel. Realising artificial photosynthesis would result in a carbon neutral source of fuel, which can be used to power homes and vehicles.

02 AR GOGGLES

Unfortunately, Google removed its Google Glass from the market, but we have seen technology demonstrations of these glasses at work, and they are incredibly useful. Microsoft has just announced the HoloLens 2 at the MWC in Barcelona. From medical applications to assembly line workers, AR Goggles have a ton of potential, and this is one of those technologies that really should be more commonly available. The challenges of power delivery, gesture recognition and real time graphics processing have already been solved. The only real roadblock is bringing the cost down.

03 CELLULAR AGRICULTURE

Cellular agriculture is a biotechnology that uses various approaches to grow animal products in lab environments. The most promising aspect of cellular agriculture is lab grown or cultured meat, that was first realised in 2013. Startups around the world are trying to bring to the market various kinds of meat, that are artificially grown using stem cells or other cells that rapidly reproduce, given a source of protein. While muscles have successfully grown in the lab, efforts to grow bone and blood vessels have failed so far. These may be achieved in the future, allowing meat to taste exactly like the "natural" versions. A lack of funds for research is the reason why lab grown meat is not already available in markets and restaurants around the world. Startups are also exploring ways to grow gelatin, eggs and fish. Other cellular agriculture efforts are focused around spinning artificial silk, or growing leather in the lab.

04 CLAYTRONICS

Claytronics uses a number of robotic claytronic atoms, or catoms that can rearrange themselves in relation to each other in three dimensions. These are swarms of nanorobots that can assume any number of shapes. Intel and Carnegie Mellon University are actively conducting research into claytronics. Currently, the demonstrations can allow the robots to self organise in two dimensions, and researchers are working to get the catoms working in three dimensions as well. Claytronics can serve a number of functions, and are 3D displays that you can hold and physically manipulate, instead of just view or control through gestures. One of the applications is called Pario, which is long distance telepresence, where two people holding a conversation can actually simulate physical interactions, with one nanobot swarm on either side. Essentially, claytronics will allow us to create the liquid metal T-1000 from Terminator 2.

05 FOOD REPLICATOR

One of the projects that Nestle is working on is a food replicator like the ones seen in Star Trek. Nestle imagines this device to be the next microwave in kitchens.